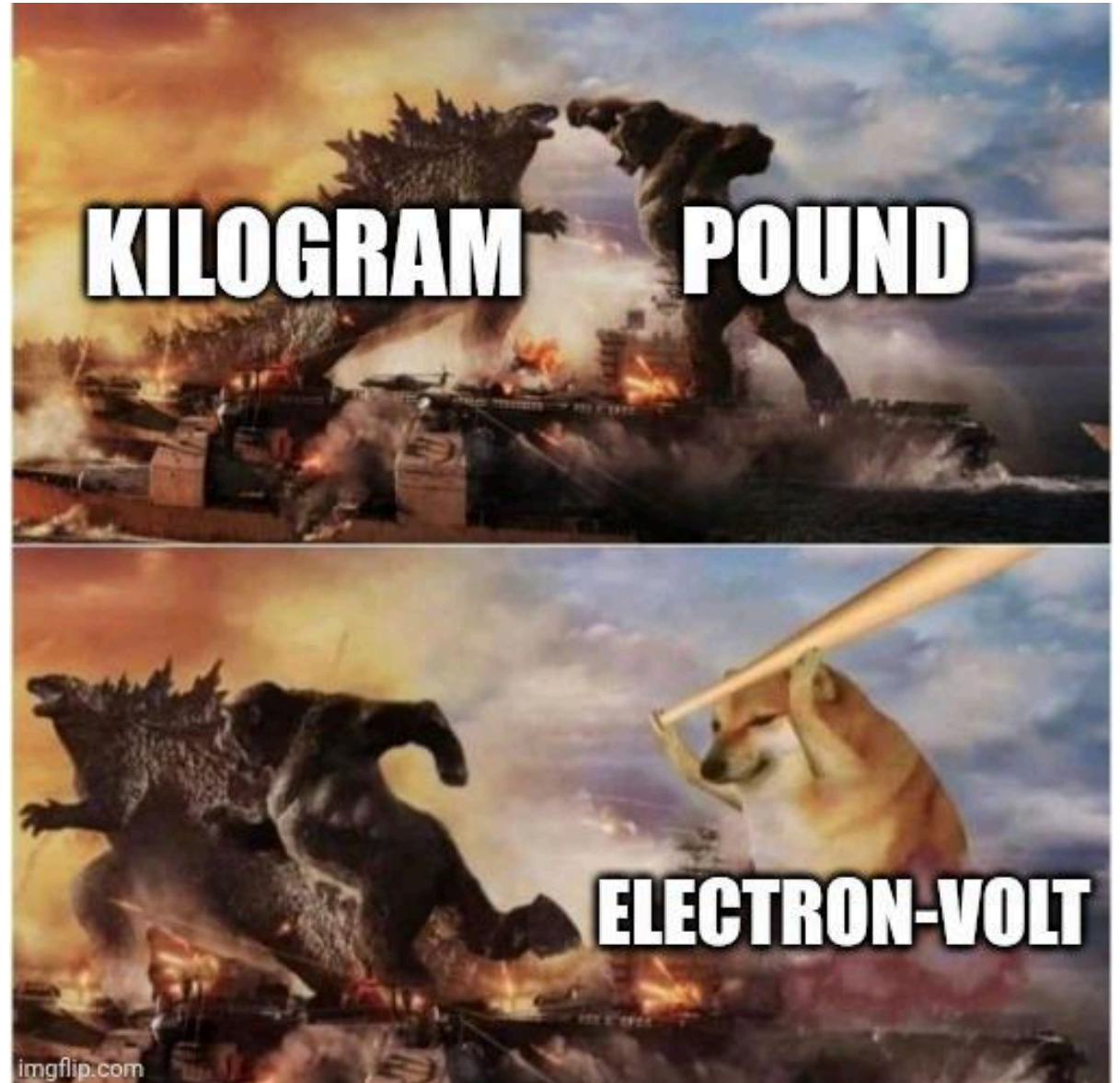


Day 08 - Workshop Day

Fridays are reserved for
your questions, homework
issues, and general
support

PHY 321 Classical Mechanics I -
Spring 2026



Announcements

- Homework 2 is due today (there is no penalty for late submission)
- Homework 3 is posted (due next Friday; late after Sunday)
- Afternoon help session (with Mihir, 3-5pm) on Zoom:
 - Zoom Link: <https://msu.zoom.us/j/99550311023>
 - password: phy321msu
- **First midterm is coming up** (assigned 16 Feb)
 - One exercise will ask you to get started on your final project planning.
 - *Who are you gonna work with? What are you interested in studying?*
Start thinking about this!

At your table, chat for 3 minutes.

What questions do you have?

👆 Make sure you upvote other's
questions.

 **RaiseMyHand**

Scan to Join Session



Point your phone camera at this QR code to join the session 3 / 8

HW 2; Exercise 5, ball thrown along a sloped ramp

A ball is thrown with initial speed v_0 up an inclined plane. The plane is inclined at an angle ϕ above the horizontal, and the ball's initial velocity is at an angle θ above the plane. Choose axes with x measured up the slope, y normal to the slope, and z across it.

- 5a: Write down Newton's second law using these axes and find the ball's position as a function of time. **Make sure to include the FBD and any assumptions you make.**

HW 2; Exercise 5, ball thrown along a sloped ramp

A ball is thrown with initial speed v_0 up an inclined plane. The plane is inclined at an angle ϕ above the horizontal, and the ball's initial velocity is at an angle θ above the plane. Choose axes with x measured up the slope, y normal to the slope, and z across it.

- 5b: Show that the ball lands a distance

$$R = 2v_0^2 \frac{\sin \theta \cos (\theta + \phi)}{g \cos^2 \phi}$$

from its launch point. **This is measured up the ramp (i.e., along it).**

HW 2; Exercise 5, ball thrown along a sloped ramp

A ball is thrown with initial speed v_0 up an inclined plane. The plane is inclined at an angle ϕ above the horizontal, and the ball's initial velocity is at an angle θ above the plane. Choose axes with x measured up the slope, y normal to the slope, and z across it.

- 5c: Show that for given v_0 and ϕ , the maximum range up the inclined plane is:

$$R_{\max} = \frac{v_0^2}{g(1 + \sin \phi)}$$

HW 3; Exercise 3, Drag force

We can observe that the models for linear and quadratic drag forces are given by:

$$f_{lin} = 3\pi\eta Dv \quad f_{quad} = \kappa\rho Av^2$$

where D is the "length scale" of the object (e.g., the diameter of the sphere), η is the viscosity of the fluid, ρ is the density of the fluid, A is the cross-sectional area of the object, and κ is a constant of order unity (larger for flat and blunt bodies; smaller for streamlined bodies).

Parts 3a and 3b

- The Reynolds number is defined as $Re = \rho v D / \eta$. What is the physical meaning of this number? For a spherical object, show that the ratio of the quadratic drag force to the linear drag force is given by $f_{quad} / f_{lin} = Re / 48$. Use this to explain the physical meaning of the Reynolds number. **Note: you may assume that $\kappa = 0.25$ for a sphere.**
- Explain a situation where there would be a low Reynolds number. What about a high Reynolds number? Estimate the Reynolds number for a falling rain drop, a parachutist, a car, and a plane.